Case Study: Kernel Optimization as a K3 Reinforcement-Learning Domain

Deep Dive into Kimi K3 — Chapter 30

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Foundation Books Project

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learned optimization skill is a claim about a policy.

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Goal: freeze the operator, choose experiments that separate hypotheses, gate the score on correctness and a whole-call timing boundary, show one exact policy update, and keep each claim at its own unit of evidence.

Roadmap

Contract before speed

From execution to a score

Learning across turns

Evidence of capability

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Freeze the operator, then diagnose

One head-wise row: $Y_{r,j} = \mathrm{sigmoid}(G_{r,j})\,\omega_j\,U_{r,j}/\sqrt{d^{-1}\sum_k U_{r,k}^2 + \varepsilon_N}$; the divisor is the head width d, even when a block pads to $p > d$ lanes.

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- Tiny row $(3, 4, 0)$: correct $\sqrt{28/3}$, wrong divisor $\sqrt{29/4}$, unmasked sentinel 5 $\sqrt{53/3}$, both $\sqrt{54/4}$. [DERIVED: exact teaching calculation]

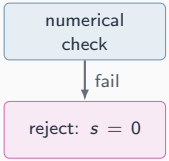
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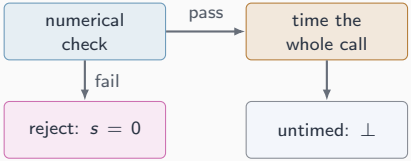
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- Chain $C_0 \to C_4$: C_1 fails only at $d = 65$; C_2 repairs the divisor; C_3, C_4 combine two or four partial sums. Python models, not GPU kernels.

From execution to a score

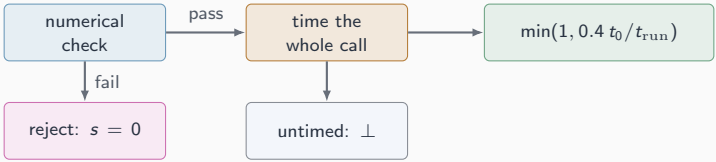
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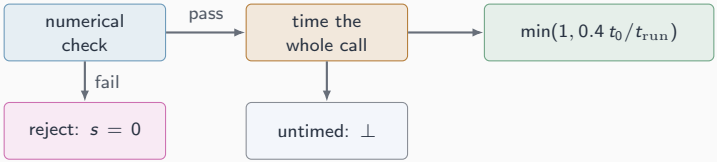


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Assigned-cost controller: C_3 selected ($Q = 0.75$); C_4 is last but scores $13/21$. Component $10\times$ faster $\Rightarrow$ whole call $\approx 1.099\times$. [DERIVED: assigned inputs, not runtimes]

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K3 reports a zero-beyond-threshold reward, 0.5 at expert performance, and shortcut checks — not an interpolation formula. [REPORTED: K3 Sec. 4.2.4]

Learning across turns

Credit the first decision with later outcomes

- $s_D = (0.4, 0.4, 0.4)$, $s_E = (0, 0.4, 1)$: first returns favor D at $\gamma = 0.5$ (0.70 vs 0.45) and E at $\gamma = 1$ (1.20 vs 1.40).

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- KernelGYM probe: returns $(5, 4, 5, 2)$ give $(0, +2, 0, -2)$; excluding A_2 before grouping flips B_2 to $+2$. [OBSERVED: controlled probe]

Evidence of capability

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One case does not isolate kernel-specific RL from test-time search. Next: Chapter 31, Agent Evaluation, Correctness, and Safety.